Avinash Agarwal

Civilization, AI, Healthspan, Space

Distributed Systems and Artificial Intelligence

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Summary

Seasoned Software Engineer with **8+ years of industry-leading experience at Meta and Microsoft**, passionate about tackling civilizational challenges, especially Healthspan and Space exploration! Proven track record of applying a research-driven approach to architecting highly available distributed systems processing billions of requests, leading cross-functional teams, and driving multi-million-dollar cost optimizations.

Experience

Meta

August 2025 — September 2026

Full-time Software Engineer (supervised by Gary G. Liu, Bill Hamilton, & Yun Zhang)

Menlo Park, California, USA

- **Applied AI & LLM Evaluations:** Drafted into the Applied AI organization to create model-evaluation tasks for Meta's next-gen LLM codenamed Avocado, under Meta's Superintelligence Labs.
- **AI for Healthspan:** Authored a project proposal within the Applied AI org to process vast amounts of genetic data; advancing milestones like Google's AlphaFold, the proposal leverages modern AI to establish novel health benchmarks and accelerate cures for aging, rare diseases, etc.
- **Engineering Productivity:** Root-caused an issue around pull-request insights where reviewer delays were incorrectly and entirely attributed to authors; filed a high-priority bug to ensure accuracy and fairness in developer analytics for company-wide engineering productivity.
- **AI Tooling & Productivity:** Championed the adoption and company-wide rollout of browser-native AI tools, accelerating company-wide productivity for everyday workflows.
- **LLM Product Direction:** Championed consistent collaborative initiatives (w/ Abhishek Yadav) to include specialized lightweight LLMs, pivoting away from computationally intensive legacy models.
- **Biometric Authentication Methods:** Ramped up on existing biometric authentication modalities (including Face ID, Touch ID, Passkeys, etc.) and contributed to engineering design reviews for their inclusion in Meta's unified login workflows (w/ Zach Yang)
- **Stability Lead**: Championed platform reliability operations for the unified authentication infrastructure, by actively triaging and resolving critical alerts, identifying and filing tasks for key SLI/SLO coverage gaps, systemic propagation delays around alert resolution, etc. (w/ Xinyi Zhang)
- **Engineering Velocity & Code Reviews:** Maintained an uninterrupted code review streak of nearly a hundred days, consistently unblocking cross-functional pull requests and upholding high review standards across core repositories.
- **WhatsApp-based Authentication:** Authored a technical proposal to prioritize secure code-send channels while simultaneously projecting millions of dollars in cost-savings; fixed critical bugs around contact eligibility (w/ Feng Wan) and internal server semantics; triaged further security improvements, video-call discrepancies, etc.
- **Meta.ai Platform Reliability:** Stabilized end-to-end test automation suites for Meta.ai login flows, significantly improving CI/CD reliability and preventing regressions from being merged into production; eliminated test flakiness by root-causing and resolving UI wait-time synchronization issues; and propagated missing test environment configuration values across cross-platform test suites, including Reality Labs frameworks.
- **On-Call Agent:** Enhanced an alert summarizer agent by resolving prompt completeness errors; collaborated directly with the company-wide on-call platform owners to include this agent in their workflow, significantly improving engineering productivity (w/ Nanway Chen)

- **On-Call Operations & Incident Management:** Completed specialized onboarding (w/ Nanway Chen); monitored dashboards, triaged alerts, completed reports and contributed to on-call investigations and reviewing fixes (w/ Feng Wan, Apurvi Saini, Brian Wang, etc.); proactively identified and root-caused a latent privacy incident (w/ Emily K. Nguyen, Nanway Chen, etc.), and collaborated with relevant tool owners to improve documentation, accelerate time-to-mitigate, and implement guardrails to prevent potential exposure in future outages.
- **Secured the runner-up position in an internal Hackathon:** Demonstrated high engineering velocity by rapidly fixing a high volume of issues while triaging improvements for the hackathon platform.
- **Dogfooding & Virtualization:** Supported on-device dogfooding for unified login flows (w/ Nanway Chen) and ramped up on internal virtualization frameworks for testing most scenarios virtually (w/ Feng Wan)
- **Unified Authentication Method Infrastructure:** Led the end-to-end engineering design and implementation of over eight distinct account-recovery eligibility methods; instrumented relevant telemetry validating the safe transitioning of billions of account recovery requests across the Meta ecosystem (w/ Zach Yang, Emily K. Nguyen, Feng Wan, Yuxi Tao, etc.)
- **Observability & Quota Management:** Audited telemetry sources to resolve persistent capacity issues; optimized sampling rates and created distinct logging partitions for partner teams (w/ Mayank Chacko)
- **Privacy-Safe Infrastructure:** Modernized legacy components to use privacy-safe wrappers while onboarding to the unified authentication infrastructure team (w/ Emily K. Nguyen)
- **Infrastructure Bootcamp:** Dedicated almost a month gaining specialized hands-on training to successfully onboard into the full time **Infrastructure Engineering** role at Meta.

Microsoft Azure Core

March 2022 — July 2025

Full-time Software Engineer (supervised by Gaurav Kapila, Chirag Gupta & Jenny Hunter)

Redmond, Washington, USA

- **Azure Resource Graph**
  - **Engineered end-to-end data ingestion components** hosted on the Azure Synapse platform to support dynamic retrieval of partitions within the specified time-windows for mission-critical security datasets (w/ Sanjai Ganesh Jeenagiri, Minerva Chen, & Umair A.)
  - **Query Optimization:** Optimized complex SQL based queries executed on Azure Synapse to efficiently load and merge concurrent incoming data streams (w/ Sanjai Ganesh Jeenagiri, Sivaraman Venkataraman, Abhinav Gupta, etc.)
  - **Ingestion Architecture Modernization:** Championed transitioning ingestion workflows from Event Hubs to Azure Service Bus, utilizing its native support for sequential ordering, deduplication windows, dead-letter isolation, etc. drastically improving data consistency across stateful data-plane operations and reducing high-volume production incidents.
  - **Partition-Key Cardinality Improvements:** Identified non-uniformity in the distribution of random subscription GUIDs across existing low-cardinality hashing functions; demonstrated expanding partition-key cardinality substantially improved partition skewness metrics and overall efficiency while reducing production incidents; reviewed custom partition selection implementations (w/ Jing Fang)
  - **High-Performance (de)serialization:** Independently root-caused bottlenecks via profiler analysis and championed the stack-wide adoption of System.Text.Json, leveraging its faster, lower-allocation APIs to optimize payload processing at scale (w/ John Call)
  - **Buffered Encryption:** Implemented buffered streaming over an existing encryption algorithm to drastically reduce active memory allocations for massive strings, while ensuring identical hash outputs and downstream compatibility (w/ Alan Yang, John Call, & Olha Tkachenko)
  - **Telemetry & Partition Observability:** Instrumented granular partition-specific metadata across Event Hub ingestion streams, substantially improving time-to-detect and root-cause analysis for relevant partition-level anomalies (w/ Olha Tkachenko)
  - **Profiled and optimized services,** identifying and eliminating bottlenecks, achieving nearly 10% improvements across compute, memory, and network efficiency.

- **On-Call Operations:** Triaged, mitigated, and root-caused live production incidents during scheduled rotations; executed manual deployment validation procedures to prevent regressions during staged rollouts; actively made multiple enhancements to internal monitors, troubleshooting-guides, etc. (w/ Pradeep Beeram, Chris Shao, Dasaradha Ramu Yalamanchili, John Call, Olha Tkachenko, Pooja Narayanan, Pinwen Su, Ben Edwards, Monica Mihailescu, Nathan Miller, Alan Yang, Youke Shen, Nicholas Bostwick, Sophia Choi, Priya Narang, Edgar Zepeda, Dylan Hartono, Neelam Tikone, Priyekant Aghi, Anna Yang, Audrey Zhu, Shilpa Sharma, Nina Potrebich, Athena Nouhi, Xinyi Cao, Liphi Gao, Venugopal Latchupatula, Mahendra Patel, etc.)

- **Azure Resource Change Analysis**
  - **Streaming Pipeline (1B+ daily notifications) Migration:** Contributed to an approved transitioning design-doc and executed foundational steps to migrate the high-throughput streaming data pipeline for Change Analysis to its dedicated scale unit (w/ Daniel Hu, Harkirat Singh & Abhinav Gupta)
  - **Change Safety Integration:** Guided partner teams on integrating with Change Analysis resources, authoring and deploying new API version specifications while reviewing inbound pull requests to ensure cross-stack architectural safety (w/ Ayaz Ali & Bharath Kalidindi)
  - **Front-End General Availability:** Worked on improving Lighthouse performance scores, deployments and test automation, completing accessibility fixes, etc. (w/ Hardhik Mallipedi, Zhonghua Bao, Julia Wang, etc.)
  - **Privacy Reviews:** Led an end-to-end internal privacy review cycle for Change Analysis, establishing data-handling and retention standards for client metadata (w/ Dharmendra Agrawal)
  - **LLM-based API Governance:** Prototyped a prompt-engineering proposal (w/ Robin Chapas) to automate the categorization of Azure Swagger API properties (security, performance, costs, etc.)
  - **Proxy Resource Change Tracking:** Contributed to the design and deployment of Change Analysis backend services for Azure proxy resources, expanding auditing coverage and closing visibility gaps across nested topologies (w/ Daniel Hu)
  - **Change Actor General Availability**: Designed and presented a comprehensive design doc on a new compliance and auditing functionality enabling customers to audit WHO initiated a change and WHICH client interface (Portal, CLI, SDK, etc.) was used to make these changes; while including additional client metadata to simultaneously meet Azure Compute's requirements; Worked on cross-team collaboration, end-to-end implementation, and overall project completion (w/ Amit Dua, Virat Satyam, Aydan Yumerefendi, & Daniel Hu)
  - **Architectural Evolution:** Championed selective caching of internal columns informed by customer query telemetry (w/ Hardhik Mallipedi) to significantly reduce active memory footprint; worked on end-to-end removal of nullable fields, ensuring data integrity and API/Swagger contract migration to significantly reduce memory allocations under strict resource constraints (w/ Jaeyoung Yoon)
  - **Partition Skewness Reduction:** Diagnosed a bug resulting in numerous zero-valued partitions, experimentally demonstrating it was caused by redundant double-partitioning; uncovered incorrect Azure Event Hubs SDK usage and resolved it by leveraging the native pre-partitioned API overload, achieving over 30% improvements in compute, memory, and network efficiency (w/ Jing Fang)
  - **Fast-Path Rollout:** Contributed to the design and staged rollout of a fast-path execution flow for Change Analysis, unlocking massive throughput improvements; authored extensive test suites to ensure correctness before full rollout (w/ Jaeyoung Yoon)
  - **Backend General Availability:** Deployed relevant services, dependencies, etc. across US and China Government clouds (w/ Robin Chapas, Pinwen Su, etc.), achieving critical compliance milestones directly supporting Azure's enterprise market positioning, well ahead of a Gartner cloud review; collaborated with authorized personnel for sovereign air-gapped cloud deployments.
  - **On-Call Operations:** Triaged, mitigated, and root-caused live production incidents during scheduled rotations and actively improved efficiency through multiple enhancements to internal monitors, troubleshooting-guides, etc. (w/ Sanjai Ganesh Jeenagiri, Abhinav Gupta, Daniel Hu, Jaeyoung Yoon, Robin Chapas, etc.)

Microsoft Azure Global

June 2018 — February 2022

Full-time SWE (supervised by Subrata Roychoudhuri, Arun Ramamurthi, & Saumya Chandra)

Hyderabad, Telangana, India

- **Microsoft Community Training** — Worked on core development of an Azure-powered learning platform incubated at Microsoft Research (MSR) India to train the next billion users; driven by a personal philosophy around education being a foundational mechanism to navigate The Great Filter.
  - **Hyperscale Platform Modernization** (supervised by Sandhya Guntreddy): End-to-end technical onboarding and platform readiness for massive-scale deployments, supporting over 10 million learners and over 1 billion course-completions (verified by Garima Kamthan)
    - **Analytics Architecture Revamp:** Authored and implemented a technical proposal for a pull-based analytics model to resolve systemic data integrity bugs. Identified and decoupled synchronous database queries, significantly improving platform performance for learner workflows.
    - **Azure Load Testing & Terraform:** Identified and merged a fix significantly improving Azure-Load-Testing's support in the multi-cloud cohort supported by Terraform.
    - **Architectural Leadership:** Championed data-store scale-out strategies including SQL Server read replicas (w/ Kinshuk Singh) and distributed caching with Redis to minimize long-term technical debt while balancing rapid iteration requirements.
    - **High-Scale Data Migration:** Architected APIs to successfully execute the migration of 160M+ course-completion records from another major cloud provider to Azure (w/ Nidhi Kothari)
    - **Throughput Optimization:** Achieved RPS improvements of up-to **20x** across core APIs and **10x** across several others; analyzed execution plans and fixed SQL queries, eliminated redundant network calls and implemented strategic caching layers (w/ Kinshuk Singh)
    - **Load Testing & Performance Profiling:** Worked on end-to-end load testing and performance profiling to ensure platform stability under massive traffic spikes (w/ Kinshuk Singh); designed and implemented scripts to locally sign JWT tokens for load-testing authentication flows.
  - **UNICEF Learning Passport** (supervised by Sandhya Guntreddy)
    - **Offline Innovation (Hackathon Lead):** Spearheaded a hackathon project (w/ Alok Agrawal) to host the entire cloud service locally on Windows machines. The platform was recognized as one of TIME's Best Inventions of 2021 for educating children in conflict and refugee zones.
    - **Global Societal Impact:** This offline-first prototype was the key technical catalyst that prompted UNICEF to launch the initiative using Microsoft Community Training for no-connectivity regions, contributing to a high-scale initiative that reached 7.4 million learners across 43 countries by 2024.
    - **Strategic Architectural Leadership:** Championed and worked on containerization and migration to cross-platform .NET Core; this move was critical for the platform's ability to support diverse hardware ecosystems (w/ Vishal Goyal & Sanmoy Ray)
    - **Operational Support & Edge Strategy:** Worked on Azure IoT Hub integration to support remote management and observability for offline learning devices; assisted in testing data-sync to ensure reliable curriculum delivery in offline settings (w/ Vishal Goyal, Kinshuk Singh, & Sanmoy Ray)
  - **Azure Marketplace Public Preview Launch**
    - **Platform Architecture & Onboarding:** Ramped up on Azure Managed Applications, and presented the deployment workflow to an Azure Global leadership audience (w/ Vishal Goyal)
    - **Automated Marketplace Deployment:** Engineered robust end-to-end deployment automation workflows to provision and configure all foundational cloud resources, services, and dependencies directly within customer Azure tenants (w/ Vishal Goyal, Brian Moore & Sukhjeevan Reddy)
    - **Azure Portal UX Templates:** Designed and implemented the complete suite of Azure Resource Manager (ARM) UI definitions and portal templates, streamlining initial parameter configuration and onboarding UX for platform customers (w/ Vishal Goyal, Gary K. & Deepak Kathayat)
    - **Architectural Leadership:** Championed Azure Lighthouse for cross-tenant management, accelerating time-to-market for all of Azure Global; aligning with the broader Azure Control Plane strategy (w/ Evan Hissey) while Azure Managed Applications (w/ Ian Hickey) was still evolving.

- **Swachh Bharat Mission**
  - **Platform Modernization & Backend Migration:** Worked under the mentorship of Aditya Agarwal transitioning the training infrastructure from Microsoft Research's (MSR) MEC platform, executing backend migrations while ensuring absolute data integrity.
  - **Cross-Functional Launch Triage:** Collaborated closely with product and program management teams (w/ Abhinav Sobti) to rapidly triage and resolve launch-critical blockers.
  - **Engineering Leadership:** Directed end-to-end engineering across core web and mobile workstreams, resolving triaged launch blockers into production-ready platform features, securing the official launch of the Swachh Bharat Mission eLearning mobile app.
  - **Xamarin UI Architecture:** Worked (w/ Venkat Kancheti, Rajeswari Chilakala, etc.) on architecting, and timely implementing a high-impact search-enabled dropdown; thus allowing users to efficiently navigate and select from the hundreds of cities where the platform was deployed.
  - **National Social Impact:** Contributed to a high-scale social impact project that successfully trained and empowered over **110,000+ municipal functionaries** across **4,000+ cities** in India.
- **Platform Search Modernization** (w/ Alok Agrawal)
  - **Unstructured Content Extraction:** Used Azure Cognitive Services to extract and process content from unstructured file formats, including documents and videos; leveraged Azure Media Services for generating transcripts, utilizing them for both search-index enrichment and subtitles.
  - **Data Ingestion & Indexing:** Engineered workflows using Azure Cognitive Search to ingest data residing across SQL and Blob Storage into a custom search index.
  - **Search Relevance & Retrieval:** Designed and implemented search relevance enhancements using pretrained semantic analyzers (edit distance, synonyms, phonetics, etc.), and a Lucene-based abstraction layer for retrieval.
  - **Deployment Automation & Cost Optimization:** Engineered deployment automation workflows to provision Azure Cognitive Search and Azure Cognitive Services within the final marketplace offering, implementing pricing tiers for search performance.
  - **Voice Search Integration:** Formulated initial architectural proposals for a voice search capability, utilizing Azure Cognitive Service's Speech-to-Text APIs to transcribe spoken queries into text.
- **Azure Front Door Integration & Performance Optimizations**
  - **Mission-Critical BCDR:** Designed and implemented a Load Balancing and Disaster Recovery architecture using Azure Front Door (already benchmarked by LinkedIn)
  - **Edge Security:** Significantly hardened the platform's security posture by configuring web application firewalls, DDoS protection, etc. at the edge.
  - **Custom Domain Configuration:** Modernized legacy custom domain configuration procedures, streamlining customer onboarding and SSL/TLS certificate management.
  - **Performance Optimization:** Benchmarked and implemented API response compression (Brotli, Gzip, Deflate, etc.) to significantly improve page-load latencies.
  - **Caching Strategy:** Implemented CDN-based caching of static assets and tuned CDN and browser caching configurations (Cache-Control) for relevant APIs.
  - **Large-Scale Migration:** Orchestrated the seamless migration of 100s of customers to the new Azure Front Door endpoints; involved careful cross-functional planning and execution.
- **Platform's Identity & Authentication Infrastructure**
  - **Multi-Region Resilience & BCDR:** Architected and implemented multi-region IdentityServer deployments behind Azure Front Door and configured active-active global load balancing and zero-downtime disaster recovery.
  - **Data Tier Decoupling & Migration:** Migrated tenant-specific deployment metadata including different configurations and token-signing certificates from SQL Server to globally replicated Azure Cosmos DB to enable low-latency global token verification (w/ Sanmoy Ray)

- **.NET Core Modernization:** Contributed to migrating the platform's core identity infrastructure including the IdentityServer components to .NET Core, while enabling container-ready execution across cross-platform environments (w/ Vishal Goyal & Saloni Chandak)
- **Auth Provider & Cost Optimization:** Spearheaded the migration of SMS based code-send provider to dedicated third-party infrastructure; collaborated directly with Telesign and engineered the end-to-end integration, drastically reducing worldwide code-send costs.

- **Platform Localization**
  - **Leadership:** Engineered a scalable internal localization framework and integrated with company-wide manual translation pipelines, eventually leading accessibility efforts across the entire platform, while keeping most of it extensible across Azure Global.
  - **RTL & Complex Scripts:** Delivered comprehensive right-to-left (RTL) format support and resolved complex ligature rendering, ensuring accurate typography across the platform.
  - **Custom PDF Engine:** Architected a templatized HTML-to-PDF certification feature within strict security and localization constraints; partnered with IronPDF to significantly accelerate time-to-market.
  - **Live Demonstration:** Presented RTL capabilities to a large audience, gracefully utilizing live product errors to demonstrate comprehensive, end-to-end localization coverage.

- **Platform Observability:** Worked on implementing a highly scalable architecture using Event Hubs and Azure Data Explorer (w/ Vishal Goyal) to aggregate telemetry across managed deployments. Utilized cross-tenant Application Insights queries and Managed Identities with AMA access to enable diagnostic and analytic dashboards (w/ Deeksha Shravani)
- **Android Build & Test Automation:** Engineered release pipelines to compile hundreds of custom APKs (w/ Sukhjeevan Reddy) in parallel; integrated with Microsoft's App Center to orchestrate automated test runs, thus expanding coverage across dozens of devices (w/ Venkat Kancheti)
- **API Test Automation:** Engineered comprehensive functional test suites using Postman, specifically scripting multi-step authentication handshakes, dynamic session token propagation, etc. seamlessly orchestrating headless execution within CI/CD pipelines via Newman.
- **Telemetry & Growth Analytics:** Integrated Application Insights with advanced funneling to track end-to-end learner journeys from landing to completion; configured a Dynamics 365 integration with Azure Marketplace to capture and generate relevant sales leads (w/ Deepak Kathayat)
- **Platform Health Probes:** Architected and implemented health-check endpoints instrumenting subsystem diagnostics for core dependencies including database connectivity, blob storage availability, certificate expiration telemetry, etc., significantly improving time to detect and mitigate outages.
- **Platform Versioning & Upgrades:** Introduced an aggregated platform versioning framework to govern system-wide release cycles; automated deployment-time execution of existing database migration scripts and relevant cross-service dependency updates.
- **Media Processing Reliability:** Resolved critical state-tracking synchronization failures across video encoding workflows by diagnosing, implementing, and rigorously testing background cron-jobs orchestrating with Azure Media Services (w/ Vishal Goyal & Anchal Goyal).
- **DRM & Content Protection:** Authored a technical proposal to modernize content protection, championing the use of industry-standard DRM protocols (Widevine, PlayReady, and FairPlay) beyond just AES-128 encryption.
- **High-Throughput Bulk Ingestion:** Architected and built an end-to-end bulk upload system enabling parallel ingestion of multi-format media; significantly reduced administrative onboarding time and super-charged the manual testing experience for everyone.
- **Dependency Graph Optimization:** Identified and eliminated cyclic dependencies across interdependent .NET project references; decoupled shared assemblies and streamlined dependency graphs, reducing overall package-size footprint **by over 70%**, drastically improving downstream workflows like build and deployment runtimes and overall developer velocity (w/ Mayukh Chakraborty)

- **Threat Modeling & Application Security:** Fixed several security vulnerabilities across both web and Android applications, utilizing both external tooling and Microsoft-internal frameworks (w/ Vishal Goyal)
- **Privacy & Compliance:** Enabled automated compliance tooling checks (w/ Mayukh Chakraborty) to identify secure and compliant dependencies; ensuring rigorous adherence to global data protection and privacy standards (w/ Vishal Goyal)
- **Deployment & Test Automation:** Engineered daily automated deployment pipelines across all internal development and test environments; orchestrated the execution of comprehensive test suites post-deployment, significantly accelerating developer velocity and overall testing productivity.
- **Offline Content & Sync Reliability:** Diagnosed and resolved critical offline content caching and learner progress synchronization issues within the Xamarin mobile application, ensuring reliable curriculum access for users in regions with zero or intermittent connectivity (w/ Kulwinder Kaur)
- **Error Diagnostics & Triage Automation:** Conducted a comprehensive deep-dive into service and client layer error handling, aggregating telemetry into Application Insights and establishing automated daily alerts prioritizing high-impact regressions.
- **State Persistence & User Analytics:** Engineered full-stack features for real-time tracking of user progress, including most-recently-viewed-lesson and last-watched-timestamp; involved REST API and database schema updates to record relevant events, and frontend enhancements across web and mobile to resume content appropriately.
- **Platform Notifications Infrastructure**: Implemented client-side Xamarin APIs (w/ Rohan Badlani) before leading the notification subsystem for the entire platform (w/ Aman Gour); supported the paginated, pull-based service and evaluated Azure Notification Hubs for real-time push notifications.
- **Mentorship & Technical Review:** Provided comprehensive technical reviews across project specifications, technical design-documents and production code for peers and mentee interns.
- **On-Call Operations:** Triaged, mitigated, and root-caused live production incidents during scheduled rotations and actively improved efficiency through multiple enhancements to internal monitors, troubleshooting-guides, etc. (w/ Aditya Agarwal, Mayukh Chakraborty, Vidhu Mangal, Anchal Goyal, Saloni Chandak, Deepak Kathayat, Abhinav Sobti, etc.)

Microsoft Garage Hackathons

June 2018 — July 2025

Engineer

Hyderabad, TN, India and Redmond, WA, USA

- **Azure Oracle:** Conceptualized a context-aware recommendation engine leveraging Change Analysis datasets to suggest next-best-actions for Azure cloud workflows (w/ Sanjeev Isaac D)
- **Kool AI(d) for Gamers:** Fine-tuned pre-trained computer vision models (w/ Shivam Tyagi) to extract real-time HUD metrics and game states from live Dota-2 streams. The collaborative team effort spanning over a hundred individuals secured 2nd place across the entire company.
- **Touchless Kiosk Ordering:** Participated in a project prototyping a computer vision gesture-recognition interface for contactless ordering on self-service kiosks (w/ Vivek Shankar, Kaustabh Kakoty, etc.)
- **Koev Halev:** Actively contributed at the IDC campus, providing domain expertise in cross-functional design discussions for a holistic project focused on mental and emotional well-being.
- **Sustainable Cloud Provisioning:** Participated in a project prototyping a solution to explicitly identify and improve visibility of carbon-efficient Azure SKUs.
- **Web3 Compute Utilization:** Participated in a project proposal to utilize idle Azure compute to support blockchain networks, providing resources to operate cryptographic mining and validation nodes.
- **Immutable Voting Ledger:** Participated in a project championing the use of cryptographic blockchain frameworks to ensure immutability and overall integrity in democratic elections.
- **IoT Home Automation:** Collaborated within a team (w/ Abhishek Raj, Shivam Tyagi, etc.) to build a Raspberry-Pi powered IoT automation layer to wirelessly control legacy electric switches retrofitted with motorized 3D-printed actuators.

Microsoft Code.Fun.Do
May 2018 — December 2019

Engineering Lead and Pizza Logistics Moderator (supervised by Vijay Vavilala & Sandeep Chadda) — India

- **Event Moderation & Algorithmic Track Proposal:** Delivered talks on technical-guidelines, project-submissions, etc; managed ad-hoc troubleshooting and Pizza logistics at NIT Surathkal and BITS Pilani (Hyderabad); championed supplementing competitive project-based events with structured problem-solving contests (Google Kickstart, Google Code Jam, Meta Hacker Cup, ACM-ICPC, etc.) positioning rigorous algorithmic proficiency as a foundational pillar for engineering talent.

Indian Institute of Technology (Banaras Hindu University)
January 2018 — May 2018

Teaching Assistant (supervised by Santwana Mukhopadhyay & Manushi Gupta) — Varanasi, Uttar Pradesh, India

- Teaching Assistant for the **Mathematical Methods** course MA-203, offered by the Department of Mathematical Sciences; evaluated and graded assessments for 500+ undergraduate students.

Goldman Sachs
September 2016 — October 2016

Software Engineering Internship Lead (w/ Shivam Tyagi & Samiksha Gupta) — Bengaluru, Karnataka, India

- **GS Quantify 2016:** Led a 3-member team to **a national runners-up position** in Goldman Sachs' flagship interdisciplinary engineering and quantitative contest.
- **Algorithmic Problem Solving:** Served as the primary technical lead and core contributor, formulating and implementing solutions for the advanced algorithms track.
- **Data Science**: Contributed to the data-science track for bond-liquidity-prediction; specifically engineering an impactful feature column, and reviewing model selection and cross-validation strategies.
- **Quantitative Finance:** Co-directed the mathematical and financial modeling track to formulate the final quantitative project deliverables.
- **Executive Presentation:** Earned a fully sponsored invitation to their Bengaluru headquarters (hosted at the prestigious Hilton Embassy) where the team delivered the concluding presentation.

Microsoft Foundry
May 2016 — July 2016

Software Engineering Internship (supervised by Pankaj Garg & Saumya Chandra) — Hyderabad, Telangana, India

- **Conversational AI:** Engineered chat-bots using the Microsoft Bot Framework, architecting custom templatized conversational flows for form-filling, within a cross-functional team (w/ Abhishek Shanthkumar, Jayasurya Jeyakodi, Akshita Rastogi, etc.)
- **Computer Vision:** Processed images using OpenCV to better highlight information, significantly improving downstream OCR engine performance (w/ Vishal Chauhan)
- **Natural Language Processing:** Evaluated experiences built with Microsoft LUIS (w/ Nikitha Gollamudi)
- **Live Technical Demonstration:** Delivered the final product presentation to a large Microsoft Foundry audience, showcasing the conversational flows for form-filling!

Xerox Research Centre
Dec 2015 — Jan 2016

Machine Learning Research Internship (supervised by Vaibhav Rajan) — Bengaluru, Karnataka, India

- **Xerox Research Innovation Challenge:** Tackled the mortality prediction problem for ICU patients using physiological vitals, subsequently combining it lab datasets to model broader clinical complications.
- **Neural-Networks & Batched-Processing:** Independently implemented and trained a custom neural network model in GNU Octave using the provided patient vitality indicators within a batch.
- **Synthetic Data & Stream-Processing:** Supplemented patient vitals with synthetic data, subsequently transitioning the neural network's training and prediction workflow from a batched-processing architecture to a continuous stream-processing architecture for near real-time predictions.

- **Winter School on Machine Learning:** Completed intensive coursework on advanced ML paradigms.
- **Data Pipeline & Ensemble Modeling:** Preprocessed massive unstructured clinical lab datasets and combined them with dynamic patient vitals; subsequently built an ensemble of ML models to predict broader downstream complications (w/ Dushyant Kumar & Sartaj Singh)
- **Final Standing:** Secured **3rd place** (online-round) and **7th place** (onsite-round) in the overall challenge; maintaining a competitive standing against specialized academic and PhD research teams.

Indian Institute of Technology (Banaras Hindu University) April 2015 — August 2015
Machine Learning Research Internship (supervised by Santwana Mukhopadhyay) Varanasi, Uttar Pradesh, India

- **DIH Project:** Completed a summer research internship funded by the Design and Innovation Hub (DIH) under the Dean of Research and Development (w/ Abhishek Bind & Arun Meena)
- **Literature Review & Formulation:** Surveyed domain literature on time-series forecasting, and search-query analytics to model and forecast regional tourism demand.
- **Data Ingestion & Preprocessing:** Aggregated historical visitor records from regional tourism board portals, paired with query volume indices extracted from Google Trends.
- **Predictive Modeling:** Developed statistical benchmarks and predictive machine learning models correlating digital search volume with physical tourist footfall to forecast tourism trends.

ACCOMPLISHMENTS

- **Google Kickstart Round D 2017:** Ranked **83rd internationally.**
- **Competitive Programming:** Reached Purple on Codeforces and Orange on CodeChef.
- **Codeforces Round #410 (Div. 2):** Ranked **15th internationally (2nd in India)**
- **DotA-2 Milestones:** Achieved a **15-match win streak** on Invoker (one of the highest skill-ceiling heros in the game), a **13-match win prediction streak**, and secured **3 Battle Cup championships**.
- **ACM-ICPC 2016** (w/ Umang Goel & Himadri Mishra): Qualified for and represented IIT (BHU) Varanasi at the Asia-Amritapuri regionals in Mysuru, Karnataka.
- **IndiaHacks 2016** (w/ Sartaj Singh): Secured a **14th rank** against 3,000+ other teams in their data-science track; invited to the on-site finals held at Vivanta by Taj in Bengaluru, Karnataka.
- **ACM-ICPC 2015** (w/ Aditya Kapoor & Shivam Tyagi): Qualified for and represented IIT (BHU) Varanasi at the Asia-Amritapuri regionals in Coimbatore, Tamil Nadu.
- **CodeChef Long Challenge:** Reached a **national rank of thirty-one** in the overall track, including a **national rank of nineteen** in the May-2015 Long Challenge.
- **HackerEarth October Sky:** Ranked **16th** among **1700 international candidates.**
- **JCECE 2013 (Medicine): State Rank 1** in the Jharkhand Combined Entrance Competitive Examination!
- **JCECE 2013 (Engineering): State Rank 2** in the Jharkhand Combined Entrance Competitive Examination!
- **NEET 2013: All India Rank 6355** among 650k+ candidates in the National Eligibility cum Entrance Test.

PROJECTS

- **Open Source contributions to AIMACode:**
  - **Google Summer of Code 2017 Project:** Authored an architectural proposal to implement pseudocode for the Probability module (Chapters 13–15) from the 4th Edition of Russell & Norvig's canonical textbook *Artificial Intelligence: A Modern Approach*.

- **Abstractions & Pseudocode Alignment:** Modernized code in the AIMA-Java repository using Java-8's functional idioms and streams, decoupling helper routines like goal-state checks and node generation, ensuring pseudocode alignment across a broad suite of search strategies.
- **Priority Queue Optimizations:** Formulated an architectural proposal to eliminate $\mathcal{O}(N)$ linear-time arbitrary node-lookup overhead in priority-queue based search frontiers, achieving $\mathcal{O}(\log N)$ time-complexity using balanced-tree data structures.
- **Ported Propositional Logic Infrastructure (C#):** Implemented missing algorithmic foundations in the AIMA-CSharp repository, adding core Abstract Syntax Tree (AST) classes to represent propositional logic, thus improving cross-ecosystem parity with AIMA Java.

- **Multi-Agent Environment Simulation:** Built a simulation of independent RL-Agents coordinating with each other to efficiently collect items distributed in a two-dimensional grid.
- **Steganography:** Implemented a system to securely embed and exchange hidden messages within digital images, enabling applications such as authenticated watermarking and media provenance.
- **The Chess Game:** Architected and developed a full-stack chess platform in Java with live multiplayer lobbies and an interactive chatroom, subsequently extending it with an AI-driven single-player mode utilizing Minimax and Alpha-Beta pruning.
- **2D Physics Game Engine:** Built a 2D rigid-body simulation in Java featuring multi-object collision detection, kinematic interactions, and configurable gravity environments; including uniform near-Earth acceleration ($\mathbf{g} \approx -9.8\,\mathrm{m/s^2}\,\hat{\mathbf{j}}$) and Newtonian inverse-square gravitational fields ($\mathbf{F} = -G\frac{m_1 m_2}{r^2}\hat{\mathbf{r}}$)
- **Tic-Tac-Toe:** Architected and developed a full-stack multiplayer Tic-Tac-Toe game in Java, subsequently extending it with an AI-driven single-player mode utilizing an if-else decision tree across two difficulty tiers, engineered to be unbeatable on hard.
- **Calendar Day Calculator:** Built a Java application to calculate the exact day of the week handling dates across millennia including logic to handle chronological edge cases.
- **CPU Overclocking**: Successfully overclocked an Intel Pentium E5200 processor from a base clock of 2.5GHz to 2.8GHz; involved tuning FSB frequencies, CPU multipliers, and voltage settings; benchmarked thermal output and system stability under peak loads.

SKILLS

- **Leadership** — Accountability, Collaboration, Prioritization, Alignment, Reviews, etc.
- **Software Engineering** — Data Structures and Algorithms, System Design, API Design, Git, CI/CD, Distributed Systems, Event-Driven Architecture & Data Pipelines (Billions of Daily Notifications), Authentication and Identity Systems, Testing and Quality Assurance, etc.
- **Artificial Intelligence** — Reinforcement Learning (Agentic Workflows, Exploration vs Exploitation tradeoffs, Play, etc.), Deep Learning (Neural Networks, CNNs, RNNs, Transfer Learning, etc.), Machine Learning (Mathematical Modeling and Search Heuristics), Data Science (Statistics, Classification, Regression, Principle Component Analysis, etc.), LLM Fundamentals and Applications, etc.
- **Networking** — TCP/UDP, Load Balancing, Reverse Proxies, CDN, HTTP/HTTPS, DNS, RPC, etc.
- **Databases & Storage** — Redis, NoSQL, SQL, OLAP/OLTP, Blob Storage, GFS/HDFS, Blockchain, etc.
- **Cloud & Infrastructure** — IaaC, Microsoft Azure, AWS, Service Fabric, Microservices, Kubernetes, etc.
- **Site Reliability** — Business Continuity and Disaster Recovery, On-call and Incident Response, etc.
- **Search & Retrieval** — Inverted Indexes, Semantic Analyzers, Apache Lucene, Elasticsearch, etc.
- **Frameworks & Libraries** — JVM, .NET, TensorFlow, Keras, Pandas, Scikit-learn, Caffe, XGBoost, OpenCV, Unity, React, ARM, Postman, IronPDF, NumPy, Apache JMeter, Selenium, etc.
- **Programming Languages** — C++, Java, C#, Python, TypeScript, PowerShell, Hack/PHP, LaTeX, etc.

- **Hardware** — Electronics, Gadgets, Device Assembly & Tuning, Firmware & Semiconductor Fundamentals.
- **Human-Computer Interaction (HCI)** — User Interface & Experience (UI/UX) Design, Accessibility (A11y), Internationalization & Localization, Typography, Digital Media Processing, Illustrations, etc.
- **Business** — Growth Analytics, Strategic Roadmapping, Optimization Techniques, etc.
- **Technical Writing** — Documentation, System Design, Incident Post-Mortems, Academic Research, etc.
- **Security** — Access Control, Rate Limiting, Cryptographic Data Integrity, Threat-Modeling, CTF-events, etc.
- **Privacy** — PII, OII, Anonymization & Pseudonymization, Cryptographic Data Masking, CCPA, GDPR, etc.
- **Natural Languages** — English and Hindi (Bilingual Proficiency)
- **People** — Model, Coach, Care, Ally!

EDUCATION

Coursera.org April 2018 — Present
Specializations and Certifications Online

- **DeepLearning.AI:** Generative AI For Software Development
- **The State University of New York:** Blockchain Basics and Smart Contracts
- **University of Alberta & Amii:** Reinforcement Learning Specialization
  - **Deep RL Lunar Lander:** As part of the capstone project, built an expected sarsa control agent featuring neural network function approximations, an Adam optimizer, experience replay, and a softmax policy to autonomously land a rocket within a demarcated landing zone.
- **DeepLearning.AI:** Deep Learning Specialization

Indian Institute of Technology (Banaras Hindu University) July 2013 — May 2018
Integrated Master's degree (5 years): Mathematics & Computing Varanasi, Uttar Pradesh, India

- **Academic Performance:** Graduated with a Cumulative Grade Point Average (CGPA) of 8.7/10.0
- **Thesis:** Deep Learning based Classification of Handwritten Mathematical Symbols.
  (supervised by Tanmoy Som)
  - **Data Augmentation:** Diagnosed class imbalances in the original HASYv2 dataset of 369 symbols, and implemented a rotation augmentation pipeline for underrepresented symbols.
  - **CNNs & Transfer Learning:** Leveraged local NVIDIA GPUs to fine-tune modern convolutional neural network architectures (DenseNet, WideResNet, MobileNet, and NASNet) on the final dataset using transfer learning methodologies.
  - **Empirical Results:** Demonstrated that resolving data quality and class distribution imbalances reduced model variance and improved convergence, validating the critical need for both manually curated and synthetic datasets to achieve state-of-the-art accuracies.
  - **Final Benchmarks:** Achieved state-of-the-art (SotA) test classification accuracy on the HASYv2 dataset, outperforming existing published baselines at the time.
  - **Thesis Defense:** Successfully defended the thesis through a detailed technical presentation covering CNN architectures, transfer-learning methodologies, and the final classification outcomes.
- **Campus Ambassador – Microsoft CODESS:** Partnered with Microsoft to champion inclusion in tech by leading the internal distribution of the CODESS launchpad program for female engineering students.
- **Technical Seminar & Curriculum Reform:** Delivered a presentation on Segment Tree fundamentals and championed its inclusion in the advanced data structures curriculum.
- **Problem Setter & Moderator, Technex–16:** Mathletics and Prayaas

- **CodeDef, Udyam–2016:** Ranked 1st in this algorithmic problem solving focused coding contest.
- **Analiticity, Technex–16:** Ranked 2nd in this campus-wide data-science contest.
- **COPS-OPC–2015:** Ranked 1st in this algorithmic problem solving focused coding contest.
- **Microsoft Code.Fun.Do:** Worked on developing a 2D top-down space shooter game using the Unity game engine (w/ Sumit Sahrawat, Arun Meena, & Shivam Tyagi)
- **Matheletics, Technex–15:** Ranked 3rd in this Project-Euler style algorithmic problem solving contest.
- **Rediscovering Darkness:** Stood 2nd in an Astronomy Quiz hosted by the Astronomy Club!
- **Treasure Hunt, KashiYatra–14:** Runner's up across all of Varanasi (w/ Abhishek Bind & Arun Meena)
- **Mod-Ex, Technex–14**: Ranked 2nd for building an open-source chess game (w/ Shivam Tyagi)
- **Robo-Soccer, Technex–13:** Runners-up (w/ Rishabh Daal, Sumit Sahrawat, Himank Airon, etc.) out of 100+ other teams at the Robotics-Club's event; conceptualized electronic circuit designs for a D-Pad controller to operate an All-Wheel-Drivetrain.
- **Event Organizer & Moderator**: Competitive tournaments for DotA-2, AoE-2, CoD-4, Civ-5, etc.

FIITJEE Forum for IIT-JEE
April 2011 — June 2013
Joint Entrance Examination
Dhanbad, Jharkhand, India

- **Stood 1st** in the batch of 2013, among approximately 400 other candidates!
- **All India Rank 1672** in IIT-JEE (ADVANCED) — 2013, among 150k candidates!
- **All India Rank 494** in Joint Entrance Examination — 2013, among 1.3 million candidates!
- Drafted into the elite **FastTrack** batch for top-performing students!

Delhi Public School
June 2011 — April 2013
Central Board of Secondary Education
Dhanbad, Jharkhand, India

- **Scored 96.5%** in the All India Senior School Certificate Examination (AISSCE) — 2013!
- **Awarded DST INSPIRE Science Scholarship for ranking in the top 1%** of AISSCE — 2013!
- Recognized in the **top 0.1% of AISSCE — 2013 for excellence in Physical Education!**
- Drafted into the **Aryabhatt** advanced mathematics program; competed in the school-level IMO round.
- **Rectifier:** Conceptualized full-wave AC-to-DC conversion and capacitive output filtering circuit designs.
- **Theory:** Mathematics, Language & Literature (English & Hindi), Physics, Chemistry, Biology, etc.
- **Practicals:** Biology, Chemistry, Physics.

De Nobili School
March 2000 — March 2011
Indian Certificate of Secondary Education
Sijua, Jharkhand, India

- **Scored 87.9%** in the Indian Certificate of Secondary Education — 2011 examination.
- **Received 90%+ in FTRE Scholarship** in a national-level talent exam among 100k+ candidates!
- **Science Exhibition Projects:**
  - **Barometer:** Built a pressure barometer in a team of 2, using a sealed jar, a flexible rubber diaphragm, and a graduated scale to track ambient atmospheric pressure variations.
  - **Water Level Controller:** Engineered an automated overflow-prevention circuit in a team of 3, leveraging tap-water conductivity and contact probes for real-time liquid-level detection.
  - **Periscope:** Built a prism-based periscope in a team of 3, using right-angle glass prisms and a cardboard-box housing to demonstrate total internal reflection and observe objects over obstacles.

- **Art & Craft Projects:**
  - **Cruise Ship Model:** Built a scale cruise ship in a team of 2 using ice-cream sticks, focusing on structural assembly, craft precision, and architectural balance.
  - **Origami Crafts:** Independently crafted geometric modular shapes, complex paper-folding models, and decorative art designs.
- **Theory:** Mathematics, Language & Literature (English & Hindi), Physics, Chemistry, Biology, Computers, Environmental Sciences, History, Civics, Geography, etc.
- **Practicals:** Reading, Speaking, Writing, Physical-Training, Art & Craft, Computers, Chemistry, Biology, Physics, Topography, etc.

PURSUITS

- **Metaverse** — Brain Computer Interfaces, Spatial Computing, Augmented and Virtual Reality, etc.
- **Science** — Medicine, Anatomy, Physics, Chemistry, etc.
- **Culinary** — Nutrition (Advanced), Foodie (Intermediate), Cooking (Beginner), etc.
- **Fitness** — Swimming (Beginner), Yoga (Beginner), Gym (Moderate), etc.
- **Sports** — RTS, FPS, CP, Chess, Badminton, Martial-Arts, etc.
- **Multiverses** — RPG, Podcasts, Documentaries, Books, Anime, etc.
- **Music (Listening)** — Rock, Pop, EDM, MeaningWave, LoFi, Rap, etc.
- **Natural Languages** — Spanish, Mandarin, French, Japanese, Tamil, Punjabi, Marwari, etc.
- **Comedy** — Writing, Humor, Satire, Irony, Wordplay, Memes, etc.
- **Philosophy** — Ontology, Causality, Epistemology, Morality, AI-alignment, etc.
- **Traveling** — Stars, SpaceX, NASA, Auroras, Universities, Museums, Monuments, etc.
- **Music (Composition)** — Lyrics, Eartuning, Staff Notation, etc.
- **Guitar** — Beginner (Electric, Yousician, etc.)